Assignment Feedback

Student: Mahmoud Hassan

Assignment: math2

Grade: 60 / 100

Feedback

Question Breakdown:

Section A (Iterated Integrals)
Max Marks: 15
Your Marks: 12
Marks Lost: 3
Reason: In Section A(iii), the student failed to provide the explicit derivation of the integration limits for the cylinder and plane intersection. Steps involving the transition from the double integral to the evaluation of bounds are insufficiently detailed.

Section B (Double Integrals Over General Regions)
Max Marks: 25
Your Marks: 16
Marks Lost: 9
Reason: Significant logical failure in B(ii): The student calculated a negative value (-2) for a volume. This indicates an incorrect setup of the upper and lower bounding surfaces. In mathematics, a scalar volume cannot be negative; this conceptual error is penalized heavily.

Section C (Double Integrals in Polar Coordinates)
Max Marks: 20
Your Marks: 10
Marks Lost: 10
Reason: Section C(ii-1) is incomplete, failing to account for the full geometric symmetry of the sphere/cylinder. Section C(iii) contains an evaluation of an improper integral that lacks the formal limit notation required at university level.

Section D (Triple Integrals)
Max Marks: 20
Your Marks: 14
Marks Lost: 6
Reason: Major arithmetic and notation mistakes in D(6) and D(7). The final result in D(6) is mathematically incoherent based on the previous integration step. D(7) lacks a simplified final form, leaving the answer as an unreduced expression.

Section E (Line Integrals and Green's Theorem)
Max Marks: 20
Your Marks: 8
Marks Lost: 12
Reason: In Section E(3), the student used a single specific path to 'show' path independence, which is logically insufficient; a formal proof via the conservative field condition was required. Section E(4) is entirely missing/crossed out, resulting in a zero for that component.

Overall Summary:
The student exhibits a basic understanding of multivariable integration but demonstrates severe lack of rigor. Calculating a negative volume and failing to formally prove path independence are catastrophic failures of logic. Furthermore, the omission of Green's Theorem suggests an incomplete mastery of the curriculum.

Major Mistakes:
- Calculation of a negative volume in Section B(ii)
- Invalid logical methodology for proving path independence in Section E(3)
- Complete omission of Green's Theorem evaluation in Section E(4)
- Incorrect integration boundaries for symmetric solids in Section C

How To Improve: